K33

TUGAS 08 MA1101

No.

Date:

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* Tutorial 3

1 25b. $\int \frac{(t+1)\sqrt{t}}{t^2}\,dt$

$= \int \frac{t\sqrt{t}+\sqrt{t}}{t^2}\,dt$

$= \int \frac{1}{\sqrt{t}} + \int \frac{1}{t\sqrt{t}}\,dt$

$= \int t^{-\frac{1}{2}} + \int t^{-\frac{3}{2}}$

$= (2\sqrt{t} + c_1) + (-2t^{-\frac{1}{2}} + c_2)$

$= \boxed{2\left(\sqrt{t} - \frac{1}{\sqrt{t}}\right) + C}$

2 25c. $\int \frac{2}{5}\sec\theta\tan\theta\,d\theta$ dari $\frac{d}{d\theta}\sec\theta = \tan\theta\sec\theta$

$= \frac{2}{5}\int \tan\theta\sec\theta\,d\theta$

$= \boxed{\frac{2}{5}\sec\theta + C}$

3 25e. $\int x^2\sqrt{x^3+4}\,dx$

misalkan $u = x^3 + 4$

$\Rightarrow du = 3x^2\,dx \Rightarrow dx = \frac{1}{3x^2}du$

$= \int x^2 \cdot \sqrt{u} \cdot \frac{1}{3x^2}\,du$

$= \frac{1}{3}\int \sqrt{u}\,du$

$= \frac{2}{9}\sqrt{u^3} + C$

$= \boxed{\frac{2}{9}\sqrt{(x^3+4)^3} + C}$

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4 25f $\int (\tan t + \sec t)^{10} . \sec t \, dt$

misalkan $u = \tan t \sec t$

$\Rightarrow du = \sec^2 t + \tan t \sec t \, dt$

$\Rightarrow du = \sec t (\tan t + \sec t) \, dt$

$\Rightarrow dt = \dfrac{1}{\sec t \, u} du$

$\int u^{10} . \sec t . \dfrac{1}{\sec t . u} du$

$= \int u^9 du$

$= \dfrac{1}{10} u^{10} + C$

$= \boxed{\dfrac{1}{10} (\tan t + \sec t)^{10} + C}$

5 26c $\dfrac{dy}{dx} = \dfrac{6x - x^3}{2y}$

$\Rightarrow 2y \, dy = (6x - x^3) dx$ .. integralkan

$\Rightarrow y^2 = 3x^2 - \frac{1}{4} x^4 + C$

$(y = 3$ dan $x = 0)$

$\Rightarrow 3^2 = 3(0)^2 - \frac{1}{4}(0)^4 + C \Rightarrow C = 9$

Dengan demikian persamaan diferensialnya adalah

$\boxed{y^2 = 3x^2 - \frac{1}{4} x^4 + 9}$

6 27b. $\dfrac{d^2 r}{dt^2} = \dfrac{2}{t^3}$; $r'(1) = 1$, $r(1) = 1$.

Integralkan didapat

$\dfrac{dr}{dt} = -\dfrac{1}{t^2} + C \Rightarrow \dfrac{dr}{dt}(1) = 1 = -\dfrac{1}{1} + C$

$\Rightarrow C = 2$

$\dfrac{dr}{dt} = -\dfrac{1}{t^2} + 2$

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Integralkan lagi didapat

$r = \frac{1}{t} + 2t + C \Rightarrow r(1) = 1 = \frac{1}{1} + 2(1) + C$

$\Rightarrow C = -2$

Dengan demikian,

$\boxed{r = \frac{1}{t} + 2t - 2}$

7 31. $h_0 = 10$ m, $v_0 = 5$ m/s, $g = -10$ m/s^2 (arah ke bawah)

(a) $h = h_0 + v(t) + a(t)$

$= 10 + 5t + \frac{1}{2}gt$

$\Rightarrow \boxed{h(t) = (10 + 5t - 5t^2)\text{ m}}$

(b) $h'(t) = 0$

$\Rightarrow 5 - 10t = 0$

$\Rightarrow \boxed{t = \frac{1}{2}\text{ sekon}}$

$h(\frac{1}{2}) = 10 + 5(\frac{1}{2}) - 5(\frac{1}{2})^2$

$= 10 + \frac{5}{2} - \frac{5}{4}$

$= \boxed{11\frac{1}{4}\text{ m}}$

(c) $h(t) = 0$

$\Rightarrow 10 + 5t - 5t^2 = 0$

$\Rightarrow 2 + t - t^2 = 0$ [Jadi $t^2 - t - 2 = 0$]

$\Rightarrow (t-2)(t+1) = 0 \Rightarrow \because t \geq 0,\ t = 2$

$h'(2) = 5 - 10(2) = \boxed{-15\text{ m/s}}$

8 33. Cara Rumus:

$v_0^2 = v_i^2 - 2as$

$\Rightarrow 0 = 20^2 - 2a.50$

$\Rightarrow \boxed{a = 4\text{ m/s}^2}$

(atau, agar konsisten,

$\boxed{a = -4\text{ m/s}^2}$).

BIG BOSS

Tutorial 4

2c. $\int_1^4 (x^2+2x)\,dx$

$\Rightarrow \Delta x = \frac{4-1}{n} = \frac{3}{n}$ | $f(x_i) = f\left(1+\frac{3i}{n}\right)$

$x_i = 1 + \Delta x . i$ | $= 3 + \frac{12i}{n} + \frac{9i^2}{n^2}$

$= 1 + \frac{3i}{n}$

$\Rightarrow \int_1^4 (x^2+2x)\,dx = \lim_{n\to\infty} \sum_{i=1}^{n} f(x_i)\,\Delta x$

$i \Rightarrow \frac{n(n+1)}{2}$ $\quad = \lim_{n\to\infty} \sum_{i=1}^{n} \left(3 + \frac{12i}{n} + \frac{9i^2}{n^2}\right)\left(\frac{3}{n}\right)$

$i^2 \Rightarrow \frac{n(n+1)(2n+1)}{6}$ $\quad = \lim_{n\to\infty} \frac{3}{n} \sum_{i=1}^{n} \left(3 + \frac{12i}{n} + \frac{9i^2}{n^2}\right)$

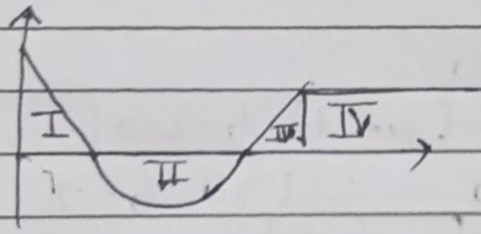
$= \lim_{n\to\infty} \frac{3}{n}\left(3n + \frac{12}{n}\left(\frac{n(n+1)}{2}\right) + \frac{9}{n^2}\left(\frac{n(n+1)(2n+1)}{6}\right)\right)$

$= 9 + 18 + 9$

$= \boxed{36}$

3d. $\int_0^6 f(x) \Rightarrow$ Luas kurva antara $x \in [0,6]$

$\Rightarrow L_1 - L_2 + L_3 + L_4$



$L_1 = 1$

$L_2 = \frac{1}{2}\pi$

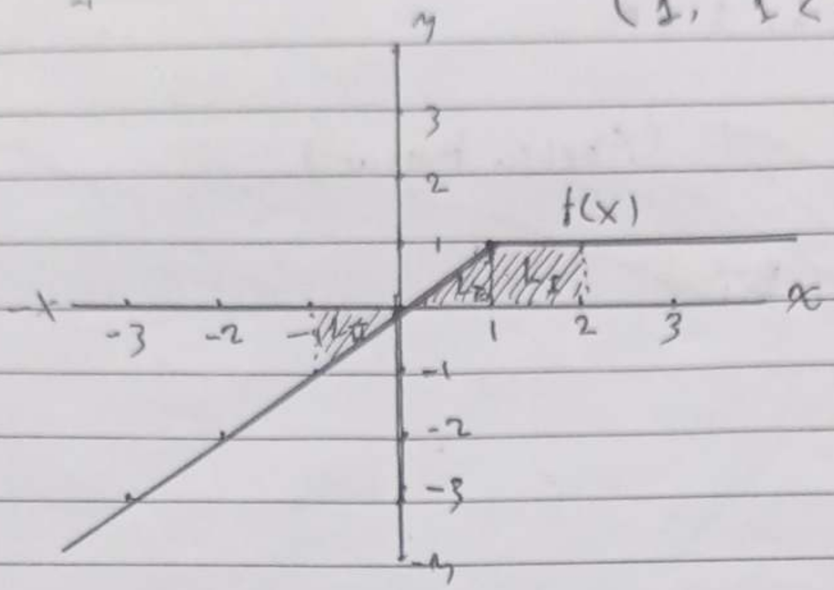
$L_3 = \frac{1}{2}$

$L_4 = 2$

Maka $\int_0^6 f(x) = 1 - \frac{1}{2}\pi + \frac{1}{2} + 2$

$= \boxed{\frac{7-\pi}{2}}$

4b. $\int_{-1}^{2} f(x)\,dx$ dari $f(x) = \begin{cases} x, & x \leq 1 \\ 1, & 1 < x \end{cases}$



$L_I + L_{II} - L_{III}$

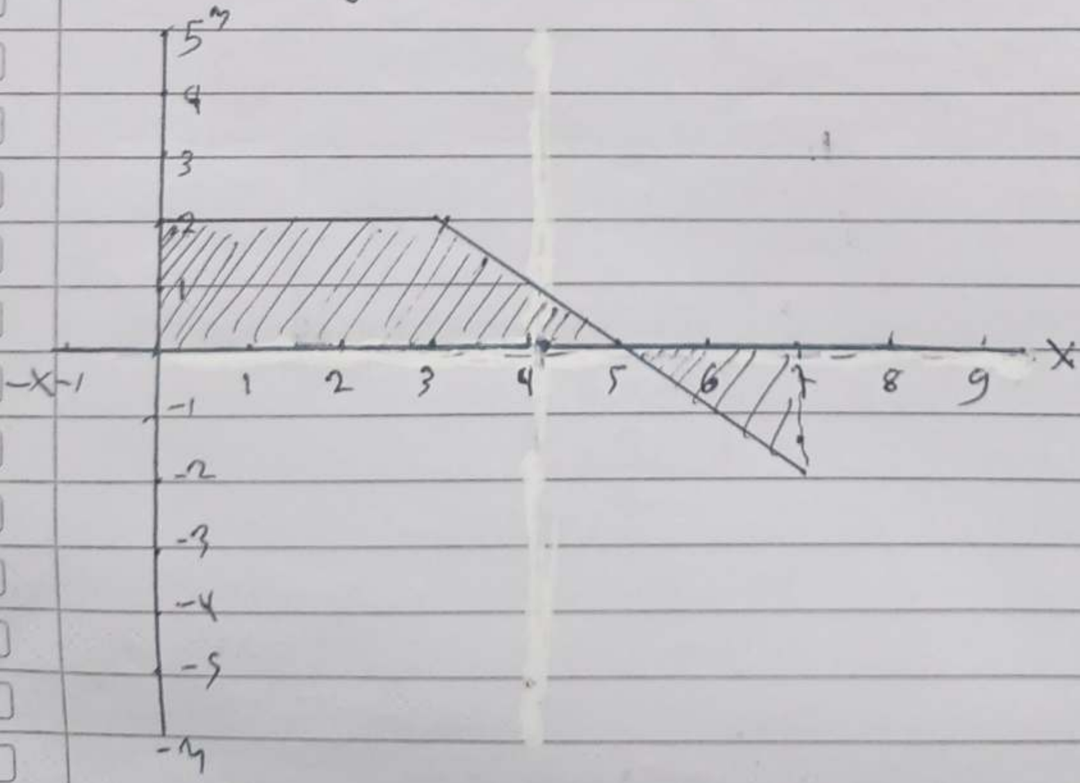
$= 1 + \frac{1}{2} - \frac{1}{2}$

$= \boxed{1}$

$\frac{}{2}$

6. $f(x) = \begin{cases} 2 & 0 \leq x < 3 \\ 5-x & 3 \leq x \leq 7 \end{cases}$

$A(x) = \int_0^x f(t)\,dt$ dengan $x \in [0,7]$.



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Formula untuk $A(x)$ dapat berupa

$$A(x) = \begin{cases} \int_0^x 2\,dt & 0 \le x < 3 \\ A(3) + \int_3^x 5 - x\,dx & 3 \le x \le 7 \end{cases} \quad \begin{matrix} = 2x - 2(0) \\ = 6 + (5x - \frac{1}{2}x^2) - (5(3) - \frac{1}{2}(3)^2) \end{matrix}$$

$$= \boxed{\begin{cases} 2x & 0 \le x < 3 \\ -\frac{9}{2} + 5x - \frac{1}{2}x^2 & 3 \le x \le 7 \end{cases}}$$

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Dapat dicek bahwa $A'(x) = f(x)$ karena kita hanya menggunakan integral biasa

$$\frac{d}{dx} A(x) = \begin{cases} \frac{d}{dx} 2x & 0 < x < 3 \\ \frac{d}{dx}\left(-\frac{9}{2} + 5x - \frac{1}{2}x^2\right) & 3 \le x \le 7 \end{cases}$$

$$= f(x) = \begin{cases} 2 & 0 \le x < 3 \\ 5 - x & 3 \le x \le 7 \end{cases}$$

sesuai dengan $\boxed{\text{Teorema Fundamental Kalkulus}}$

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7d. $\frac{d}{dx} \int_x^{2x+1} \cos(\theta^2)\,d\theta$, misalkan $F(\theta) = \int \cos(\theta^2)\,d\theta$

$\Rightarrow \frac{d}{d\theta} F(\theta) = \cos(\theta^2)$

$$= \frac{d}{dx} F(\theta) \Big|_x^{2x+1}$$

$$= \frac{d}{dx}\left(F(2x+1) - F(x)\right)$$

$$= 2F'(2x+1) - F'(x)$$

$$= \boxed{2\cos((2x+1)^2) - \cos(x^2)}$$

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~~8 Kemonotonan ⇒ turunan pertama~~

$\cancel{\frac{d}{dx} F(x) = \frac{d}{dx} \int_0^{2x-4} \frac{t}{9+t^2}\,dt}$

$\cancel{= \frac{2t}{9+t^2}}$

~~Pembuat nol $\Leftrightarrow 2t = 0 \Rightarrow t = 0$, maka pembaginya~~

~~$(-\infty, 0)$ dan $(0, \infty)$~~

~~$(-\infty, 0)$, ambil $t = -1 \Rightarrow -\frac{2}{5} < 0$ (turun)~~

~~$(0, \infty)$, ambil $t = 1 \Rightarrow \frac{2}{5} > 0$ (naik)~~

8. $F(x) = \int_0^{2x-4} \frac{1}{4+t^2}\, dt$

$\Rightarrow F'(x) = 2\left(\frac{(2x-4)}{4+(2x-4)^2}\right)$

$= \frac{4x-8}{4x^2-16x+20}$

$= \frac{x-2}{x^2-4x+5}$

Kemonotonan ketika pembilang $= 0$ sebagai titik absis kritis atau tak terdefinisi

$\Rightarrow \frac{x-2}{x^2-4x+5} = 0 \Rightarrow x = 2$

penyebut pasti taknol karena $D < 0$

$(-\infty, 2)$ ambil $x = 0 \Rightarrow -\frac{2}{5} < 0$ (turun)

$(2, \infty)$ ambil $x = 3 \Rightarrow \frac{1}{2} > 0$ (naik)

Dengan demikian, selang kemonotonannya adalah

monoton naik $= (2, \infty)$	dengan titik absis kritis
turun $= (-\infty, 2)$	pada $\boxed{x = 2}$

Kecekungan menggunakan turunan kedua

$\Rightarrow F''(x) = \frac{(x^2-4x+5) - (x-2)(2x-4)}{(x^2-4x+5)^2}$

$\Rightarrow F''(x) = \frac{-x^2+4x-3}{(x^2-4x+5)^2} = 0 \Rightarrow x^2-4x+3 = 0$

$(x-1)(x-3) = 0$

maka titik belok absisnya adalah $\boxed{x \in \{1, 3\}}$

Cek kecekungan

$(-\infty, 1)$, ambil $x = 0 \Rightarrow \frac{-3}{(\dots)^2} < 0$ (bawah)

$(1, 3)$, ambil $x = 2 \Rightarrow \frac{1}{(\dots)} > 0$ (atas)

$(3, \infty)$, ambil $x = 4 \Rightarrow \frac{-3}{(\dots)} < 0$ (bawah)

Dengan demikian, selang kecekungannya adalah

cekung ke bawah $= (-\infty, 1) \cup (3, \infty)$

cekung ke atas $= (1, 3)$

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9b $\int_0^{2x} f(t)\, dt = 8x^2 + \sin(6x)$

$F(2x) - F(0) = 8x^2 + \sin(6x)$ turunkan

$2f(2x) = 16x + 6\cos 6x$

$\Rightarrow f(2x) = 8x + 3\cos 6x$

$\Rightarrow f(x) = 4x + 3\cos 3x$

Dengan demikian,

$f(t) = 4t + 3\cos 3t$

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